

Improving computer-mediated information alignment in production organizations

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Abstract This paper describes a developing project to describe and quantify concepts of information alignment between computerized information systems and other aspects of information sharing in a production facility. Major elements of this project were components of the first author's experience in an operational production organization, including the work to transform industrial practice into opportunities for new research. A critical aspect of information alignment, as developed in this framework, is the recognition that computer-based information systems must be implemented with an eye to effective integration with other forms of information exchange and production knowledge in the organization.

Keywords Information networks · Production management · Information alignment · Team performance

Introduction

Graduate students receive a lot of advice while in graduate school. At some point during graduate school that advice is transcribed into lessons that can lead to the success of that student as a researcher. One important lesson is how to critique literature; another is how to formulate a paper to demonstrate a need for new research. One of the more critical lessons is how to identify the right question to answer. Lastly, in order to enjoy doing research the most valuable lesson is how to tailor the research to satisfy personal interests.

Those students interested in industry jobs normally tailor their research toward interests or experiences they had on various job assignments.

Gavriel Salvendy, PhD taught a research seminar at Purdue University tailored to teach students how to critique literature and develop research from the literature. The focus of each critique was the transferability from a global standpoint and the implications of the future of the research. There were discussions of what was done well, what could have been done better, who would be interested in the research. That course taught three of the four important lessons discussed above. The lead author took one of Dr. Salvendy's seminars in the Spring of 2006, and worked for a government contractor that built guidance and navigation units (GNUs) for various missiles, aircrafts, and land navigation. While working for the company, SAP and eFactory were being implemented throughout the site and production floor, respectively. The author witnessed the difference in meeting contract deadlines and productivity during the transition to becoming paperless during three internship stints over three consecutive years. Therefore, this paper focuses on improving computer-mediated information alignment in production organizations.

Multiple issues affect aligning information systems in a manner that improves upon the performance prior to the implementation of the system. There are issues with how operators interact with computers and the complexity of the interface. How organizations tailor their organization to information systems, instead of the other way around, which changes processes that have been documented. There are also issues with the approach to decide on an information system. This paper also addresses how to quantify the effect of information alignment between legacy systems (i.e. meetings, excel documents, paper reports, etc.) and information systems (i.e. eFactory, SAP, Oracle, etc.).

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Organizations and performance

The prevalence of work teams in organizations highlights the need to understand the factors that influence group task performance (Whitney 1994). Although there are innumerable types of groups operating in organizations today, including autonomous work groups, project groups, and multifunction work teams, these groups share one common characteristic: They exist for pursuing some objective or goal (O'Leary-Kelly et al. 1994). Differences among work teams pose an immediate, practical challenge for management. Some teams, such as those in production and service, tend to be synchronized with counterpart work units and customers. Others, such as design and development teams, have missions calling for creativity and innovation. They may need special help in applying group processes to their resources. Team managers need to be sensitive to such differences when making decisions on such issues as team training and consultation, physical environments, performance measurements and feedback systems, rewards systems, and other contextual features (Sundstrom et al. 1990).

Just having a collection of people working on a specific set of tasks does not make an organization successful. What makes an organization successful are the dynamics of that collection of people, and the properly set goals that are achievable. These goals are the baseline of performance measures. The interaction and inner workings of teams are defined as group dynamics, which are the interacting forces within a group concerned with the sociological study of these forces.

Sundstrom et al. (1990) define performance measures as the acceptable output to customers within or outside the organization who receive team products, services, information, decisions, or performance events (such as presentations or competitions). In the field of engineering these cross-functional teams, along with their interactions and activities, are defined as concurrent engineering which is defined as the consideration of all the contributing activities likely to affect the company's product of service life cycle or quality. These teams are directed by the organization, which assigns goals that are strategic in defining success. These goals are identified as performance measures and the team's progress toward that metric is called effectiveness. Effectiveness is defined as the capability of, or success in, achieving a given goal.

In pursuit of goal attainment, it is understood that group members should be motivated to improve their performance plans because group members understand that more efficient work methods contribute to improved group performance (Weldon et al., 1991). The proper management of group performance, effectiveness, and impact weigh heavily on goal setting and achievement in order to obtain profitable organization success.

These organizational of managing performance and effectiveness lead to a primary emphasis on organizational and production alignment between goals, resources, demands, and constraints. Alignment can be seen as an essential definitional and measurement question for the organization. Gladstein (1984) identifies four very intriguing questions that begin to evaluate group performance, which are questions about alignment:

1. What kinds of activities need to go on across group activities?
2. Under what conditions are these activities needed?
3. Do these activities relate to performance?
4. How do intragroup processes and boundary activities relate to one another?

Defining information alignment

Alignment research can be broken down into four categories (performance, process, scheduling, and information). Performance alignment is the vertical and lateral alignment of project, team, department, business segment, global sector, and organization performance measures. Each entity has its own goals, responsibilities, and performance measure of success. Linking all that to the organization's performance measures is called performance alignment. Process alignment is similar to the International Organization for Standardization (ISO) certification meaning that the organization is operating as documented. Process alignment is difficult to assess when considering varying levels of control. Control is defined as the amount of intervention (or other response activities not documented as general process tasks) allowed to workers to return the system to a stable state. Stable state is defined as system operation that is documented or predicted (excluding off-nominal events). The level of process alignment is determined by how many or how well processes map to the organization's documentation and actual organization activity.

A number of researchers have been concerned with the resources necessary to operate an organization. In the case of resource allocation, that type of alignment falls under scheduling. Scheduling alignment is a lateral analysis of how resources are shared across units and over time. Priority logistics or scheduling algorithms dictate the sharing of resources. Information alignment spans all areas of operation and decision making, and is essential to knowledge of system state. Misalignment may be defined as any instance where an organizational requirement is not addressed by information technology or organization operations and policies (Soh et al. 2003). With misaligned information, decision makers have an inaccurate representation of the system state and resource

requirements. Decisions made utilizing inaccurate representations make the environment vulnerable to errors that may evolve into adverse events. Among other pertinent factors, information alignment is vertical, lateral, and time sensitive.

This study is concerned with developing quantifiable measures of information alignment using empirical data from various environments. [Giachetti \(2004\)](#) raised the issue of “pertinent information” at high and low levels of coordination. “Pertinent information” is defined as the right information at the right time in the right format. Organization alignment involves aligning many different levels of an organization and its business segments to allow more accurate predictability of organization performance. To align an organization requires consideration of many different levels of vertical and lateral alignment of processes, resources, and information. In past research, this pertinent information alignment is usually described structurally, such as which general processes are in place to document and store information. However, there is a growing need, particularly in time critical industries, to consider pertinent information needs dynamically—whether such information is available to actors who need it within time constraints specified by organizational and situational demands.

Vertical alignment is critical from the perspective of coordinating ideas and requests along the organization hierarchy. Vertical alignment is defined as the upward and downward mapping of alignment factors along the organization reporting structure or organization chart. Lateral alignment is defined as the mapping of alignment factors across the organization chart. Alignment is essential in order to provide the proper representation of the system state.

When coordinating organization interaction, one must consider the integration process in terms of who needs to interact, in what capacity, and at what frequency. [McCarthy \(2006\)](#) outlined several essential components that numerous enterprise architecture frameworks directly or indirectly addressed. The components addressed were alignment, integration, value creation, change management, and compliance. Along with [McCarthy \(2006\)](#), this study is concerned with alignment and integration with change management becoming important for future research. Although the components proposed by [McCarthy \(2006\)](#) are difficult to track, they do provide a logical sense of what is needed as the value of information in an organization is concerned.

In distributed teams, alignment is used to connect the flow of resources and information. The type of information or resources required by each team to perform their duties may widely differ. The information and resource need differences raise the concerns of how to manage the generation of information, flow of information, and scheduling of resources when all sources of information do not come from a centralized source. Alignment does not simply mean centralization, but the creation, distribution, and identification of informa-

tion, knowledge, or resources at the proper phase, frequency, and in the appropriate format to improve reaction time and productivity.

To make alignment easier to improve, organizations may investigate situation awareness, tools, procedures, policies, and processes, among other organization operation aspects. When investigating organization awareness from a micro level, workers can only be aware up to a certain threshold even with training. Information technology as a tool is normally used to increase situation awareness because implemented algorithms embedded in the software can decipher the system state and display the information needed in the format necessary to make decisions based on the system’s actual behavior. An organization can evaluate other tools or resources to determine if the function, operation, or scheduling can be improved to a certain specification to improve organization effectiveness and efficiency. Procedure and policies may be amended to remove inefficiencies by giving more control to users when off-nominal events occur. Irrelevant processes may be removed, and processes that foster favorable interactions within the organization and with other organizations may be added. However, effective alignment still requires that these amendments occur before integration failures or a lack of awareness leads to catastrophic outcomes.

Information for individuals and organizations

[Caldwell \(1997\)](#) stated that the examination of individual performance is necessary but not sufficient for understanding performance at the group or organization level. When moving from the individual to the group or organization level there are significant additional components required to facilitate effective performance ([Caldwell 1997](#)). The issue with group, team, or organization alignment is there are more connects, overlaps, and confounds than when considering individual performance.

One function of within-team communications is to provide members with information about what their teammates are doing because at times decisions and outcomes depend on each member knowing the status of the others’ tasks ([Fussell et al. 1998](#)). This criterion of awareness should also be factored into information alignment because suboptimal situation awareness (SA) and information alignment result in poor performance. It is generally accepted that if SA is suboptimal than adverse consequences occur ([Gorman et al. 2007](#)). The key is measuring and recovering from suboptimal conditions by monitoring conditions with understandable and unintrusive measures.

[Fiore et al. \(2003\)](#) discussed improved team productivity as a coordination space of team interaction over time and distance. To coordinate over time and distance, a technological intervention is required. The use of traditional technologies

such as email among team members may affect both coordination and overload (Fussel et al. 1998). Gorman et al. (2007) analyzed email traffic to assess situation awareness for critical versus non-critical events. Even though certain decisions require one to be passively aware of team member's activities, overload in this case occurs when an email is not viewed or not viewed in time due to an excess in demands. When moving from a structural to a dynamic analysis, the issues of time, medium of coordination, and workload have been introduced as critical factors of group performance.

Research has shown that to develop effective team work measures a framework or model of team performance needs to be constructed (Undre et al., 2007). This research plans to examine the effects of the alignment of inputs (information array) with the outputs (output performance measure). If this were explained from the perspective of the signal theory model, alignment is unknown in the black box (receiver) and the output is contingent upon how well processes and information work together within the team to make a valid decision that has not decayed because the input no longer describes the system state.

Effective information sharing in the presence of teams requires the active distribution of descriptions of current and future information needs to both local teammates and to the larger organization (Burstein and Diller 2004). The most critical aspect of distributed teams is sensemaking (the processing of information to make a decision). If the alignment of information and resources is poor, it is difficult to assess the system state in the time required, let alone make decisions properly and with effective integration of information. If the alignment does not allow time for sensemaking, distributed teams must deviate from suitable decisions based on facts to decisions based on incomplete and varying intuitive knowledge bases. This reliance on unshared, nonaligned information can cause unfavorable results (Stasser et al. 2000).

Accurate measures of situation awareness (SA) should be primary indicators of organization performance, but many traditional measures of SA disrupt the flow of work (Gorman et al. 2007). The traditional SA measures are poorly suited for dynamic analysis in operational organizations, because they disrupt normal behavior and skew the analysis of organization performance (Gorman et al. 2007). There is a need for the development of improved and even new parameters of SA, sensemaking and alignment. The future of information alignment research is the use of simulation to obtain predictable measurement with high ecological validity.

If information alignment research continues without the development of scalable measures, the operations of organizations will continue to have unexplainable effects of poor performance. Information alignment effects are sensitive, dynamic and ever-changing due to the many interactions, information need, and decisions made in an organization or team on a daily basis.

Technology and alignment

Maintaining awareness of others' activities is more difficult and requires some degree of technological intervention (Fussel et al., 1998). Many different perspectives exist pertaining to the use of technology to manage an organization's internal interactions. The technology used (typically an IT system) must be able to identify and diagnose the current system state just as well or better than the organization without technology. An organization is capable of doing this without technology, but if an organization is to rely on technology to carry the burden, the organization must be able to amend the technology configuration(s) to capture the operational evolution across the lifecycle of the organization. Operational evolution is defined by the fact that over time established and well-mapped processes become obsolete and provide a misrepresentation of the system behavior. Technology or the organization must be able to accommodate the change otherwise improper decisions are made in accordance to the system's behavior.

The evolution of technology, specifically production systems, has placed greater emphasis on the modeling, monitoring, and maintaining of information and resource flows to improve production efficiency. Several generations of information systems have matured to support this function; the current versions of such systems are widely known as Enterprise Resource Planning (ERP) systems. However, poor implementation and use of such systems still compromise organizational performance, in large part due to the inadequate alignment between the information system capabilities and the operational dynamics of the production system.

Improving old processes by simply applying information technology ignores organization infrastructure, processes, and performance deficiencies (Luftman et al. 1993). Many corporations implement information technology with the intent to improve organization performance. However, the answer is not information technology, nor is it passively including group members. The proper means to improve organization performance from an information standpoint is to properly diagnose the organization's information needs, which starts with the evaluation of the organization's information alignment status. Once the information alignment status is understood then the proper application of an information system will assist in achieving efficient and effective organization performance as long as the organization's change management policy allows for the alignment of information technology and organization information to stay in line.

Ho et al.'s (2007) general guidelines (e.g. flexible presence, highly predictable and reliable, and support exchange of viewpoints) for designing computerized tools to support information transfer provide implied proof that information technology is not the solution to information alignment. The underlying issue has to do with what occurs before

information technology is implemented and it is obvious that to save money it must be done right the first time.

It is known that the strategic use of information technology can alter the basic nature of an organization Luftman et al. (1993). Enterprise Resource Planning (ERP) systems are one of the most popular information systems of corporations to date. Unfortunately, ERP projects report an unusually high failure rate, sometimes jeopardizing the core operations of the organization (Hong and Kim 2002). In short, ERP research is included in alignment research because technology dependent organizations and paperless initiatives have adopted these types of systems along with the confound variables affecting performance due to how well the system was implemented and how it fits with the dynamic of the organization. Al-Mashari (2002) identified a gap in ERP research by stating that research in the ERP area is still lacking and the gap in the ERP literature is vast. Individual studies that have been performed represent a taxonomy of ERP concepts and approaches. It is important to understand how an organization is operating when implementing an ERP system because research has not developed a concrete method of selection, use, analysis, and management to ensure the success of the implementation or organization performance.

If strategic alignment is only measured at the organization level, information systems and business executives might simply know that their organization was misaligned, but would not have sufficient information to isolate the source of the misalignment (Tallon and Kenneth 1998). Soh et al. (2003) alluded that misalignments often occur as a result of organizations attempting to fit their current structure to a pre-designed ERP software package. Deploying an enterprise system does not by itself create integrated and seamless organization operation (Grant 2003). Kim et al. (2005) proposed three critical impediments of ERP implementation being functional coordination, business process change for project management, and resistance of users. Chang (2006) identified that future research in information technology (IT) management should focus on the ERP implementation process and addressing issues across a broader range of organizational IT sophistication. ERP systems are not going to be the future of organization information and resource management. Therefore, it is essential to prepare organizations for the transition and that preparation is a high measure of alignment. Now the questions become: (1) what are high and low levels of alignment, (2) what is the threshold for those levels, and (3) how is an organization's performance affected in these levels?

Process contributions to alignment

There is much interest in alignment and application of process improvement philosophies in production systems literature. Kaplan and Duchon (1988) discussed qualitative and

quantitative approaches in a case study of a new information system pointing out four methodological points two of which were (1) the need for context-specific measure of job characteristics and (2) the importance of process measures when evaluating information systems. Process level performance measures are likely to yield greater insight into where the organization is misaligned, helping to isolate bottlenecks in the organization (Tallon and Kenneth 1998). Alignment tends to be difficult in the earlier stages of the product life-cycle. However, alignment achieved during early lifecycle stages is cheaper and less intrusive to product success in later earlier stages. In accordance with the production equation, significant improvement is achieved by alterations to setup. In a product lifecycle (e.g. design, product integration, and operation), the design and product integration phases are considered setup. Adopting process-level perspective would allow one to represent strategy as a series of activities or routines within each business process (Tallon and Kenneth 1998).

When significant improvements cannot be achieved by increasing resources or altering the scheduling of resources, significant improvements may be achieved by improving the alignment of the organization. From a productivity standpoint, alignment can be improved by matching information and resource needs in varying combinations across the organization. From an organization standpoint, one must determine which aspects of the organization should be decentralized and those that must be centralized. Significant improvements are achieved through the buy-in of the technology required to centralize or decentralize aspects of the organization. From an operations standpoint, when implementing the technology necessary to improve alignment, process and policy verification and validation must occur to determine if the organization is operating as anticipated from the documentation. Without aligning operation documentation with the actual operation activity, all other attempts to improve alignment will be in accordance to an incorrect system representation. When improving alignment, there needs to be a change management program implemented to prevent the actual operation activity from relapsing due to resistance and process difficulty imposed by the improvement.

Tools for measuring and improving information alignment

Fussel et al.'s (1998) Management Game pointed out how different organizations or individuals may be interested in different performance and status measures. It is important to be careful of what level whether vertical or lateral the data collection is coordinated. What may be important from a performance standpoint in one department at one site in a certain organization for management may not be the same

for operators, engineers, accountants, or business associates in the same organization let alone a different organization environment.

Fussel et al. (1998) developed two simple awareness tools (an email archiving system and an activity monitoring tool) designed to provide passive awareness of others' activities without creating information overload. The tool design reiterates the issue of sensemaking in reference to time available for decision making. Sensemaking is essential to information alignment because even if the information flows perfectly, if the information provided does not allow the recipient to make a decision before the environment changes, then the decision is decoupled from the current system state and the organization behavior is unpredictable. The activity monitoring tool would be good in assessing information alignment, but the tool fails to provide any information about receipt of information at the necessary time. Further, such a tool as commonly implemented does not consider if the environment state has changed between input (information receipt) and outcome (decision). In such conditions, it is difficult to monitor the affect of misalignment of information flow and improved information alignment.

Issues like workflow changes, complexity of data entry, inappropriate reports, and inadequate revenue computation functionality represent potential misalignments that require links to data sources and linking documents regarding those issues (Soh et al. 2003). Gorman et al. (2007) introduced a software tool titled ChainMaster that implemented the ProNet algorithm that extended the software by performing tests for existence of issues at multiple lags where multiple-link paths can be highlighted. The ability to quantify alignment may be done in multiple methods such as surveys, awareness tools including time, and mapping techniques of archive data. The authors are developing an analysis tool that utilizes archive data to quantify the effect of information alignment in various production environments. Such a tool incorporates the following elements:

1. Information flow chart
2. Information overlap chart
3. Information evaluation table categorized by collaboration stages and cognitive processes
4. Correlation analysis
5. Analysis of variance

The information flow chart is intended for use in two different formats for each environment investigated. Both type of charts were used by Palmer (2005) in a preliminary study of information alignment in a production systems environment, and will be described in more detail below. The first chart was titled an Organizational Information Reporting Flow Network (Fig. 1) and the other chart was titled Production Decision Information Flow Network (Fig. 2).

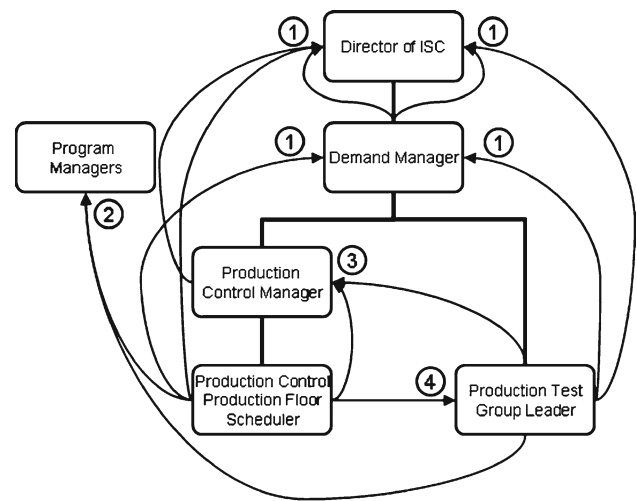


Fig. 1 Organizational information reporting flow network from Palmer (2005)

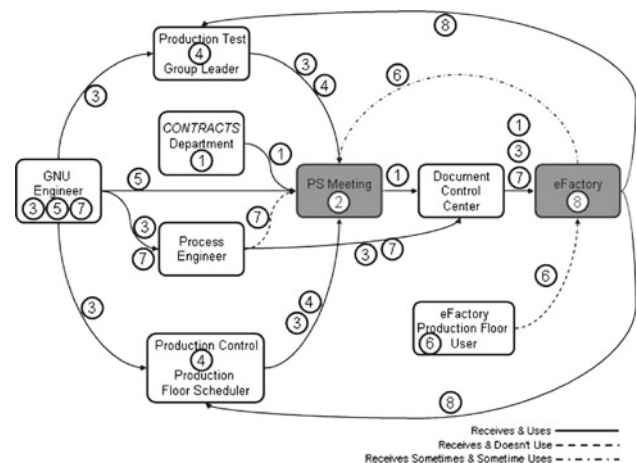


Fig. 2 Production decision information flow network from Palmer (2005)

Each box in both Figs. 1 and 2 represents the organizational entities that use information in the environment investigated. (These are presented for information flow relationships only, not for detailed understanding of those specific entities.) The circled numbers represent the information artifacts that each organization entity used, generated, and reported. Because these artifacts existed at different levels of organizational activity, this multi-level analysis must be supported in alignment considerations. Figure 3, also from Palmer (2005), displayed alignment in a manner of information usage to make decisions, which also must be approached differently. The different lines in the legends will be applied different values for Figs. 1, 2 and 3.

Ho et al. (2007) introduced the information overlap chart that connected entities with information used. Figure 4, based on Ho's chart, reiterates the fact that information can be at different levels. The information going across was in reference

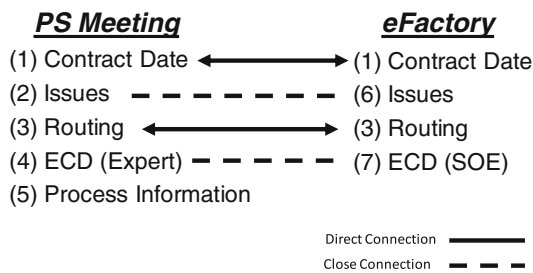


Fig. 3 Production status meeting and eFactory information alignment, from [Palmer \(2005\)](#)

	official progress notes	computer on wheels	pre-round flowsheet	daily goals sheet	unit census printout	bedside patient charts
Attending Physician						
Specialty Physician						
Resident Physician						
Nurse Practitioner						
Registered Nurse						
Medical Student						

Fig. 4 Information usage overlap chart from [Ho et al. \(2007\)](#)

to documents used to make decisions and system status evaluation.

Each entity used different information from the documents. If Fig. 4 is broken down further in the same manner as Fig. 1 through 3, a more detailed analysis can be performed making the use of Information Evaluation table more valuable. The intent for Fig. 4 is to apply a value to information usage against organization entities. Each information artifact in Fig. 4 has three attributes (generated, reported, and used) for an organization entity, each of which will be applied a value (e.g. 10, 5, and 1, respectively). The Information Evaluation table qualitatively evaluates each information artifact against the organization's performance measures in addition to another table that evaluates the organization performance measures against the actual organization output productivity. The information artifact portion of the table is organized by collaborative stages and cognitive process (e.g. Knowledge Construction, Collaborative Team Problem Solving, Team Consensus, and Outcome Evaluation and Revision), described above as important criteria for individual and organizational information alignment. The Information Evaluation table for the information artifacts against performance measures is similar to that from Fig. 4, except the numbers in the body of the table correspond to percentage that each information artifact affects the performance measures multiplied by the value determined from the flow charts and alignment chart (e.g. Figs. 1, 2 and 3 for the information artifacts).

A practical example

[Palmer \(2005\)](#) investigated an organization that classified as High Occurrence, Low Volume, and High Risk. The information flow was tracked specifically in the Integrated Supply Chain (ISC). The departments that had part in producing the organization output product was as follows: (1) Contracts, (2) Engineering, (3) Integrated Supply Chain, (4) Programs, (5) Quality, and (6) Supply. Each of the six departments attended a production status meeting that was held to provide status on production, updates on issues, and assign priority to products to ensure daily goal was met. The meeting involved the review of three documents (Progress to Plan, Rolling Action Item List, and Shortage Report). The three documents provided insight into which departments were responsible for certain information artifacts. However, most of the three document information would be used for the awareness aspect of information alignment. The information artifacts used for the Information flow chart was attained from information necessary to make production decision in the production status meeting along with the information loaded in the information technology designed to schedule production on the shop floor.

The approach is to map the information flow among organization entities. Figure 2 displays a map of the organization of where the author worked. Once the information flow is mapped the flow is given values in the information overlap table. Table 1 identifies the information overlap table with the values assigned to the following variables:

- Connection values (between legacy system and information system)
 - Direct connection (1), close connection (10), and no connection (30)
- Information type
 - Generated (10), reported (5), and used (1)
- Entity level
 - Operator/IT (5), Engineer/leader (10), and management/departments (30)
- Information usage
 - Received and used (1), Received and doesn't use (5), and receives sometimes and sometimes uses (10)

After the information overlap table, the following evaluation tables are (1) Information Artifacts against Production Performance, (2) Organization Entities against Production Performance, (3) Performance Measures against Production Performance (4) Information Artifacts against Performance Measures then Performance Measures against Production

Table 1 Updated company A information overlap table

Information artifacts	(1) Contract date					(2) Issues					(3) Product routing					(4) ECD (Expert)					(5) Process information					(6) Issues				
	10	5	1	1	1	100	50	10	10	10	5	1	1	100	50	10	100	50	10	300	150	30	100	50	10	100				
Organization entities	G	R	U	U	U	G	R	U	U	U	G	R	U	G	R	U	G	R	U	G	R	U	G	R	U	G				
Product engineer	10	0	0	0	0	0	0	0	100	50	0	0	0	0	0	0	0	0	0	3,000	1,500	0	0	0	0	1,000				
Production test group leader	10	0	0	0	0	0	0	0	0	50	10	0	0	1,000	500	0	0	0	0	0	0	0	0	0	0	0				
Contracts department	30	300	150	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
Process engineer	10	0	0	0	0	0	0	0	0	50	10	0	0	0	0	0	0	0	0	0	0	300	0	0	0	0				
Production floor scheduler	5	0	0	0	0	0	0	0	0	25	5	0	0	500	250	0	0	0	0	0	0	0	0	0	0	0				
PS meeting	5	0	0	5	500	0	50	0	0	0	5	0	0	0	0	50	0	0	150	0	0	150	0	0	50	0				
Document control center	30	0	150	30	0	0	0	0	0	150	30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
eFactory production floor user	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	500	250	0	0				
eFactory	5	0	0	5	500	0	0	0	0	0	5	0	0	1,500	750	50	0	0	0	0	0	450	0	250	500	0				
Total	300	300	40	40	500	0	50	100	325	65	0	0	0	1,500	750	50	0	0	0	3,000	1,500	450	500	500	550	1,000				

Information Artifacts	Organization Entities	(1) Contract Date			(2) Issues			(3) Product Routing			(4) ECD (Expert)			(5) Process Information			(6) Issues			(7) ECD (SOE)			(8) Queue Time			Total
		10	5	1	100	50	10	10	5	1	100	50	10	300	150	30	100	50	10	100	50	10	300	150	30	
		G	R	U	G	R	U	G	R	U	G	R	U	G	R	U	G	R	U	G	R	U	G	R	U	
GNU Engineer	10																									2
Production Test Group Leader	10																									
Contracts Department	30																									
Process Engineer	10																									
Production Floor Scheduler	5																									
PS Meeting	5																									
Document Control Center	30																									
eFactory Production Floor User	5																									
eFactory	5																									
Total																										

Fig. 6 Practical example information overlap table

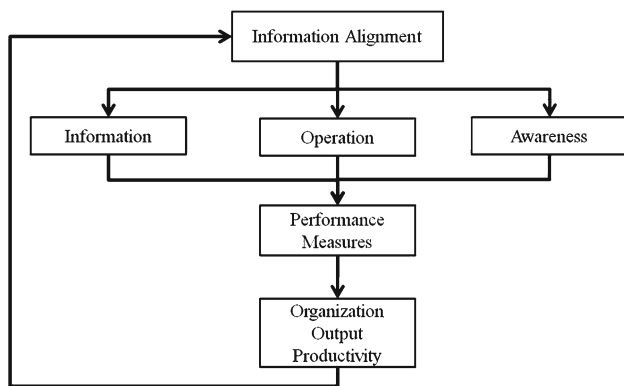


Fig. 7 Information alignment conceptual model

productivity and performance measures. Three different aspects (information, operation, awareness) of information alignment are used and an analysis will be performed. Each of the three analyses will be performed without adding the additional values from Figure 5 to understand if the additions from Table 1 produce a better model. It may show true that in certain environments the information, operation, and awareness analyses do not provide a better model than evaluation without the additional values created by Table 1. Additionally, it may show true that only one method or a combination of methods (information, operation, or awareness) provide a reasonable model to measure the affect of information alignment in an organization.

The framework and analysis techniques presented in this paper help to focus attention on the dynamic issues of information alignment, and how the media of information flow (meetings, planning systems, or other information technol-

ogies) work to support or discourage effective matching of information needs and timing to when and how information is delivered. Clearly, this is an improvement over prior analyses that simply describe the existence of an enterprise resource system as the critical element for efficient production, rather than a focused and continual examination of how the system is used, what data are reliably updated, and on what time frames those data are available to decision makers and task performers in the organization.

There is a concern that “good” and “bad” information alignment will not be determined by the sample size used in each environment. The matrix size may create an issue with conducting an analysis of variance across environments utilizing the information, operation, and awareness methods. The research may be constrained to various levels of the evaluation tables discussed after Table 1 in the Practical Example section. Therefore, the results may provide a limitation into the claims that can be made about the affect the information alignment aspects have on various environments.

A more substantial criticism that could be directed at the information alignment discussion presented in this paper is that it is limited to a particular type of production facility, or manufacturing segment. As of this writing, the authors are applying the methods described above to examine sources of information misalignment in university student services operations. A focus on a high importance student services activity (student recruiting) with respect to a time critical event (financial aid deadlines) ensures that the issues of information alignment are seen as having a similar level of impact and relevance for the university as the organization being addressed. Preliminary results of this information alignment analysis has already uncovered important gaps in information

sharing and task coordination between families, university admissions offices, and interested programs in highlighting recruiting interest to targeted students in order to provide appropriate financial aid packages. Continuing research will help to provide more effective quantitative measures of information alignment appropriate to this type of service sector activity.

The sensitivity of alignment measures across conditions has not yet been determined. If multiple organizations operate within similar constraints and performance regions, the question becomes one of determining a threshold of performance that will not allow greater differentiation of alignment. A valid comparison is defined as a range of values of information alignment effects that are similar due to the level of information alignment within the same performance region. It has yet to be determined if there is a significant difference between the information alignment effect when an organization product is produced in multiple sites in contrary to one site. Such a question is beyond the scope of this current research.

Once the alignment analysis is complete for a given organizational unit, the results of this method can determine the significant relationships and correlations that influence an organization's output and performance measures. The significant factors will provide a model that can predict some of the variance of an organization's output affected by information alignment. The variance can then be connected to a percentage of organization output. Once the correlation is made, then the future of information alignment research will be to determine the threshold of information alignment and how much of an organization's output is affected within the levels of "good" and "bad" information alignment.

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